

MATH 114 - Computational Stochastics

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1 Overview

1.1 Introduction

Stochastic models describe real life phenomena like motion of the cell, shape of polymers stock prices, and more. What can we compute analytically and numerically from these models?

Key questions:

1. What is the distribution of the system?
2. How can we sample from the distribution?

3. How can we compute mean, variance, and higher moments of these systems?

Main topics:

1. Sampling RVs
2. Markov chain Monte Carlo (MCMC)
3. Stochastic differential equations (numerical & computation)

1.2 Sampling

- Generating "hard" RVs from "easy" RVs.
- Compute "hard" integrals.

ex Area of the unit circle, $A = \pi$

Let $U_1 \sim U(-1, 1)$ and $U_2 \sim U(-1, 1)$

Let $x = \begin{cases} 0 & \text{if } (u_1, u_2) \text{ outside circle} \\ 1 & \text{if } (u_1, u_2) \text{ inside circle} \end{cases}$

$\implies P(X = 1) = \frac{A}{4}, P(X = 0) = 1 - \frac{A}{4}$

$\mathbb{E}(X) = \frac{A}{4}$

After N trials we generate $X_1, X_2, \dots, X_n$ iid

By the strong law of large numbers

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N X_i = \frac{A}{4} = \frac{\pi}{4}$$

1.3 MCMC

Construct Markov chains to generate target distribution.

ex Ising model

2 Markov Chains

2.1 Lecture 9

Def: A stochastic process is a family of RVs, $X_t \in S$, indexed by t , where $S \subseteq \mathbb{R}^d$ is a state space. Typically, $t \in [0, \infty)$, or $t \in [a, b]$, or $t \in \{0, 1, 2, 3, \dots\}$, etc.

- A discrete time stochastic process, is a sequence of RVs, $X_n \in S$, $n = 0, 1, 2, \dots$ or $n = 1, 2, 3, \dots$
- If S is finite or countably infinite then the stochastic process is a discrete state process.

First, we will examine discrete-time, discrete pace, stochastic process.

Two basic questions:

- Find the joint distribution

$$P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) \\ i_0, i_1, \dots, i_n \in S$$

- Find the long-time behavior of the distribution of X_n :

$$P(x_n = i), i \in S, n \gg 1$$

Def: Markov processes

- A stochastic process $\{X_n\}_{n=0}^{\infty}$ is a Markov process if

$$P(X_{n+1} = i_{n+1} | x_0 = i_0, \dots, x_n = i_n) \\ = P(X_{n+1} = i_{n+1} | X_n = i_n) \\ \forall n \geq 0, \forall i_0, \dots, i_{n+1} \in S$$

- A Markov process is time homogeneous if:

$$P(x_{n+1} = j | x_n = i) \\ = P(X_1 = j | X_0 = i) \\ \forall n \geq 0, \forall i, j \in S$$

Here we consider discrete time/state, time homogeneous Markov Chains (MC).

Def: Transition probability

$$P(i, j) = P(X_1 = j | X_0 = i) \\ = P(i \rightarrow j) \quad \forall i, j \in S$$

Def: Transition matrix

$$P = [p(i, j)] \text{ if } S = \{1, 2, 3, \dots\} \text{ or } \{0, 1, 2, \dots\}$$

Here i, j are states and also are row/column labels

For any $n \geq 1, i_0, \dots, i_n \in S$, we have:

$$\begin{aligned} P(X_0 = i_0, \dots, X_n = i_n) \\ &= P(X_0 = i_0)P(X_1 = i_1|X_0 = i_0)P(X_2 = i_2|X_1 = i_1, X_0 = i_0) \\ &\dots P(X_n = i_n|X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \quad [\text{conditional probability}] \end{aligned}$$

ex Let $X_0, X_1, \dots, X_n \in S$ iid. Then $\{X_n\}_{n=0}^\infty$ is a time homogeneous MC.
Let $S = \{1, 2, \dots, N\}$. Then $p(i, j) = \mathbb{P}(X_1 = j|X_0 = i) = \mathbb{P}(X_1 = j) = p_j$

$$\Rightarrow p = \begin{bmatrix} p_1 & \dots & p_n \\ \vdots & \ddots & \vdots \\ p_1 & \dots & p_n \end{bmatrix}, \sum_{i=1}^n p_i = 1$$

2.2 Lecture 10

Recap:

- Discrete time/state, time homogeneous
- MC: $\{X_n\}_{n=0}^\infty \in S$
- $\mathbb{P}(X_{n+1} = i_{n+1}|X_0 = i_0, \dots, X_n = i_n) = \mathbb{P}(X_{n+1} = i_{n+1}|X_n = i_n)$
 $= p(i_n, i_{n+1})$
- Transition probability $\mathbb{P}(X_1 = j|X_0 = i)$
 $= p(i, j) = p_{ij} \quad \forall i, j \in S$
- Transition matrix $\mathbb{P}[p(i, j)]$

ex Two-state MC, e.g. state of phone $X_n \in S = \{0, 1\}$

$$\begin{cases} X_n = 0 & \text{phone free at time } n \\ X_n = 1 & \text{phone busy at time } n \end{cases}$$

$$\mathbb{P}(\text{free} \rightarrow \text{busy in time step}) = p$$

$$\mathbb{P}(\text{busy} \rightarrow \text{free in time step}) = q$$

$$\mathbb{P}(\text{free} \rightarrow \text{free in time step}) = 1 - p$$

$$\mathbb{P}(\text{busy} \rightarrow \text{busy in time step}) = 1 - q$$

Transition matrix:

$$P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$$

$$\text{e.g. } p = \frac{1}{2}, q = \frac{1}{3}$$

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

$$\text{Let } \mathbb{P}(X_0 = 0) = \frac{1}{5}, \mathbb{P}(X_0 = 1) = \frac{4}{5}$$

$$\text{Then } \mathbb{P}(X_0 = 0, X_1 = 1, X_2 = 0)$$

$$= \mathbb{P}(X_0 = 0) \cdot p(0, 1) \cdot p(1, 0)$$

$$= \frac{1}{5} \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{30}$$

$$\text{And } \mathbb{P}(X_0 = 0, X_2 = 0) = \sum_{i=0}^1 \mathbb{P}(X_0 = 0, X_1 = i, X_2 = 0)$$

$$\boxed{\text{ex}} \quad P_0 = I, p_0(i, j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$P_1 = P, p_1(i, j) = p(i, j)$$

The distribution vector π_n for X_n is

$$\pi_n = (\pi_n(0), \pi_n(1), \dots)$$

$$\text{where } \pi_n(i) = \mathbb{P}(X_n = i), i \in S$$

Note that π_n is a probability vector

Property: Let $X_n \in S = \{1, 2, 3, \dots\}, n \geq 0$ be the time homogeneous Markov Chain with transition matrix P . Then

- P is a stochastic matrix

$$\bullet \quad P \cdot \mathbb{I} = \mathbb{I} \text{ where } \mathbb{I} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

- 1 is an eigenvalue of P with eigenvector $\mathbb{I}$

- $P_{m+n} = P_m \cdot P_n$ (Chapman-Kolmogorov equation)
- $P_n = P^n$
- $\pi_n = \pi_0 P_n = \pi_0 P^n$
- $\pi_{n+1} = \pi_n P, \forall n \geq 0$

Proof: direct verification

e.g. $\mathbb{P}(X_1 = j) = \sum_{i \in S} \mathbb{P}(X_0 = i) \mathbb{P}(X_1 = j | X_0 = i)$

$\pi_1(j) = \sum \pi_0(j) P_{ij}$

Long-run behavior, invariant distribution

Def: Let $P = [p(i, j)]$ be the transition matrix for Markov Chain $\{X_n\}_{n=0}^\infty$ with $S = \{1, 2, \dots, N\}$. A probability vector $\pi \in \mathbb{R}^n$ is a stationary or invariant distribution if $\pi^P = \pi$

2.3 Lecture 11

- Recap: Discrete time/state, time homogeneous Markov Chains. Transition matrix $P = [p(i, j)]$, $p(i, j) = \mathbb{P}(X_1 = j | X_0 = i)$, $p_n(i, j) = \mathbb{P}(X_n = j | X_0 = i)$
- Markov Chains $\rightarrow$ MCMC

ex 3 state MC

$$P = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{8} & \frac{2}{3} & \frac{5}{24} \\ 0 & \frac{5}{6} & \frac{1}{6} \end{bmatrix}$$

$$S = \{1, 2, 3\}$$

$$\mathbb{P}(X_1 = 1 | X_0 = 1) = \frac{3}{4}, \text{etc.}$$

$$\lim_{n \rightarrow \infty} P^n = \begin{bmatrix} \pi \\ \pi \\ \pi \end{bmatrix}, \quad \pi = \left(\frac{2}{11}, \frac{4}{11}, \frac{5}{11} \right)$$

$$\begin{aligned} \text{Set } \pi_0 &= (\mathbb{P}(X_0 = 1), \mathbb{P}(X_0 = 2), \mathbb{P}(X_0 = 3)) \\ &= \mathbb{P}(X_0 = i) \end{aligned}$$

Then

$$\begin{aligned}
 \pi_n &= \pi_0 p^n \\
 \lim_{n \rightarrow \infty} \pi_n &= \pi_0 \lim_{n \rightarrow \infty} p^n \\
 &= \pi_0 \begin{bmatrix} \pi \\ \pi \\ \pi \end{bmatrix} \\
 &= \left(\pi(1) \cdot \sum_i \pi_0(i), \pi(2) \cdot \sum_i \pi_0(i), \dots \right) \\
 &= \pi
 \end{aligned}$$

Remarks

- $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = j) = \pi(j) \forall j$
-

$$\begin{aligned}
 \pi_n &= \pi_{n-1} P \\
 \lim_{n \rightarrow \infty} \pi_n &= \lim_{n \rightarrow \infty} \pi_{n-1} P \\
 \pi &= \pi P
 \end{aligned}$$

- π is a left eigenvector of P with eigenvalue 1.

Def: Let $P = [p(i, j)]$ be the transition matrix for MC $\{X_n\}_{n=0}^\infty$ with $S = \{1, \dots, N\}$. A probability vector $\pi \in \mathbb{R}^n$ is a stationary or invariant distribution for X_n (or P) if: $\pi = \pi P$

Remarks

Above example shows $\lim_{n \rightarrow \infty} \pi_n =$ invariant distribution (with some conditions).

Questions: invariant distribution.

- When do they exist?
- Are they unique?
- Roughly: typically have unique invariant distribution unless reducible or periodic.

Def: A finite state time-homogeneous MC $X_n (n \geq 0)$ with transition matrix P is regular if there exists an integer $k \geq 1$, such that $P^k > 0$, i.e. all entries of P^k are strictly > 0 .

Thm

Let $X_n (n \geq 0)$ be a regular N-state Markov Chain with transition matrix P . Then:

1. $\{X_n\}_{n=0}^{\infty}$ has a unique stationary distribution π , $\pi > 0$
2. $\lim_{n \rightarrow \infty} P^n = \begin{bmatrix} \pi \\ \vdots \\ \pi \end{bmatrix}$, i.e. $\lim_{n \rightarrow \infty} p_n(i, j) = \pi(j)$
3. For any initial distribution π_0 , $\lim_{n \rightarrow \infty} \pi_0 P^n = \pi$ i.e. $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = j) = \pi(j) \forall j$

Remarks

- 2. $\implies \lim_{n \rightarrow \infty} \mathbb{P}(X_n = j | X_0 = i) = \pi(j)$ independent of i
- From 2.:

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} p_k(i, j) &= \lim_{n \rightarrow \infty} p_n(i, j) = \pi(j) \\ \text{And } \frac{1}{n} \sum_{k=0}^{n-1} p_k(i, j) &= \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{P}(X_k = j | X_0 = i) \\ &= \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{E}[\mathbb{I}[X_k = j] | X_0 = i] \\ &= \mathbb{E} \left[\frac{1}{n} \sum_{k=0}^{n-1} \mathbb{I}(X_k = j) | X_0 = i \right] \\ &\rightarrow \pi(j) \text{ as } n \rightarrow \infty \end{aligned}$$

Which means the mean fraction of time spent in $J \rightarrow \pi(j)$

How do we find π ?

- Solve $\pi P = \pi$
- Diagonalize $P = U\Lambda U^{-1}$

$$\begin{aligned}\Rightarrow \lim_{n \rightarrow \infty} P^n &= \lim_{n \rightarrow \infty} (U\Lambda U^{-1})^n \\ &= U \left(\lim_{n \rightarrow \infty} \Lambda^n \right) U^{-1} \\ &= \begin{bmatrix} \pi \\ \vdots \\ \pi \end{bmatrix}\end{aligned}$$

2.4 Lecture 12

Recap:

- Discrete time/space MC $\{X_n\}_{n=0}^\infty, X_n \in S$
- $P = [p(i, j)], p(i, j) = \mathbb{P}(X_1 = j | X_0 = i)$
- $\pi_0 = \mathbb{P}(X_0 = i), \pi_n = \mathbb{P}(X_n = i)$ (pmf X_n)
- Stationary distribution π s.t. $\pi P = \pi$
- Regular: $\exists k \geq 1$ s.t. $P^k > 0 \Rightarrow$ unique stationary distribution $\pi > 0$

$$\text{And } \lim_{n \rightarrow \infty} P^n = \begin{bmatrix} \pi \\ \vdots \\ \pi \end{bmatrix}, \lim_{n \rightarrow \infty} \pi_n = \pi$$

- Detailed balance: probability vector λ s.t. $\lambda_i p(i, j) = \lambda_j p(j, i) \forall i, j \in S$
 $\Rightarrow \lambda P = \lambda \Rightarrow \lambda$ stationary with respect to P .

Remarks

1. Can view MC as evolution of probability vectors (pmfs): $\pi_0, \pi_1, \dots$ with $\pi_{n+1} = \pi_n P$. This is hard to compute for $|S|$ large.
2. MC is a random sequence of states $X_0, X_1, \dots \in S$ which is often easier to compute.

Recall: Discrete Inversion (Lecture 6)

e.g.: Generate discrete RV X

$$P(X = i) = p_i, \quad i = 1, 2, 3$$
$$p_1 + p_2 + p_3 = 1$$

1. $U \sim U[0, 1]$
2. $k = \min\{m \mid \sum_{i=1}^m p_i \geq U\}$
3. Set $X \leftarrow k$

Algorithm: Generate / simulate MC

Input: transition matrix P , initial distribution π_0

Output: MC $X_0, X_1, \dots \in S, X_n \sim \pi_k = \pi_0 P^k$

1. Generate $X_0 \sim \pi_0$
2. For $k = 1, \dots, n$, generate X_{n+1} from $p(X_k)$. i.e. pmf over S , given by the X_k 'th row of P .
3. Repeat n times to get samples from π_k

MCMC: (R&K ch.6)

- Key idea: generate distribution π by simulating MC with stationary distribution π
- Often π is known up to a normalizing constant.
- $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \mathbb{I}\{X_k = i\} = \pi(i) \quad \forall i \in S$
- $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n g(X_k) = \mathbb{E}_\pi(g(x))$

We assume that

- RV $X \in S$ is finite, e.g. $S = \{1, 2, \dots, N\}$
- Target pmf of X is $\pi = (\pi(1), \dots, \pi_N), \pi(i) = \mathbb{P}(X = i) \quad \forall i \in S$
- π has special form $\pi(i) = \frac{1}{z} b(i) > 0 \quad \forall i \in S$.
Where $b(i) > 0$ are known and $z > 0$ is unknown norm.

$$\sum_{i \in S} \pi(i) = 1 \iff \sum_{i \in S} b(i) = z$$

Remarks

- z is hard to compute, (sum of $N = |S|$ terms)
- z is called a partition function

Algorithm: Metropolis

- First construct proposal transition matrix $Q = [q(i, j)]$, where Q is symmetric.
- Choose initial state $X_0 \in S$ (e.g. randomly).

Given $X_k = i \in S$

1. Sample $Y \in S$ from $q(i, \cdot)$, i.e. $\mathbb{P}(Y = y | X_k = i) = q(i, y) \forall y \in S$
Assume $Y = j$ is observed.
2. Let $\alpha = \alpha_{ij} = \min\left(1, \frac{\pi(j)}{\pi(i)}\right) \in [0, 1]$ and generate $U \sim U[0, 1]$.
If $U \leq \alpha$, accept Y , $X_{k+1} \leftarrow Y$
If $U > \alpha$, reject Y , $X_{k+1} \leftarrow X_k$
(Accept Y with probability α)
Output: $X_0, X_1, X_2, \dots$

Remarks

- $\{X_n\}_{n=0}^\infty$ is MC with $\lim_{n \rightarrow \infty} X_n \sim \pi$
- Can choose $Q = [q(i, j)]$ to be simple, e.g. uniform jumps, $q(i, j) = \frac{1}{N}$
- Only need the ratios $\frac{\pi(j)}{\pi(i)}$, so we don't need z .
- Call Y a proposed state.
- Call α the acceptance probability.
- In practice, $\alpha = a_{ij}$ is easy to compute.
- Can be extended to non-symmetric Q

ex 1D Ising model

$$\begin{aligned} S &= \{\sigma = (\sigma_1, \dots, \sigma_m) | \sigma_i = -1 \text{ or } +1, 1 \leq i \leq m\} \\ &= \{-1, +1\}^m \\ |S| &= 2^m \text{ states} \end{aligned}$$

2.5 Lecture 13

Metropolis Algorithm

$$S = \{1, 2, \dots, N\}$$

Goal: Sample random variable X from some target distribution $\Pi(i) = \mathbb{P}(X = i) \implies \Pi_i(i) = \frac{1}{z}b(i)$

We have $Q = [q(i, j)]$ which is our transition matrix (symmetric).

1. Choose $X_0 \in S$. Given $X_n \in S$, sample $Y \in S$ from $q(X_n, i)$, i.e.
 $\mathbb{P}(Y = y | X_n = i)$
2. Set $\alpha = \min(1, \frac{\pi(y)}{\pi(X_n)}) \in [0, 1]$.
Generate $U \sim U[0, 1]$
If $U \leq \alpha$, accept $Y : X_{n+1} \leftarrow Y$
Otherwise, $U > \alpha$, reject Y , $X_{n+1} \leftarrow X_n$

1D Ising Model

$S = \{\sigma = \{\sigma_1, \sigma_2, \dots, \sigma_n\} : \sigma_j = \pm 1\}$
 $\Rightarrow 2^m$ states.

$$\begin{aligned}\Pi(\sigma) &= \frac{1}{z} e^{-\beta \mathcal{H}(\sigma)} \\ \mathcal{H} &= -J \sum_{j=1}^{m-1} \sigma_j \sigma_{j+1} - h \sum_{j=1}^m \sigma_j \\ J, h &> 0\end{aligned}$$

Large $\Pi(\sigma) \longleftrightarrow$ Low energy states.
Calculate physical quantities.

$$\begin{aligned}m(\sigma) &= \langle \sigma \rangle = \frac{1}{m} (N_+ - N_-) \\ &= \frac{1}{m} \sum (N_+ - N_-) \Pi_p(\sigma) \\ \frac{\Pi_\beta(y)}{\Pi_\beta(X_n)} &= e^{-\beta(\Delta \mathcal{H})}\end{aligned}$$

a. $1 < i < m$

$$\Delta \mathcal{H} = 2J\sigma_i(\sigma_{i+1} + \sigma_{i-1}) + 2h\sigma_i$$